

Bayesian Inference of F1 Driver Ability

STAT 405: Bayesian Statistics - Sohbat Sandhu (79661179)

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1 Introduction

Formula One (F1) performance is shaped by a complex interaction between driver decision-making, car performance, tire degradation, and race conditions. Modern telemetry makes these interactions measurable at high resolution: instead of analyzing only lap time, we can study micro-sector time, where a lap is segmented into 100 equal-distance regions. A central challenge is that observed micro-sector times are not direct measurements of driver skill. They confound skill with team/car advantages, tire compound and wear, temperature, and race interruptions. The goal is therefore a Bayesian inference problem: to separate these factors and quantify uncertainty in driver ability.

1.1 Research Question

Using telemetry and lap data from the 2023 Abu Dhabi Grand Prix, we aim to infer latent driver ability and rank drivers

1.2 Literature Review

It is common to use telemetry-based analytics for motorsport performance, but much of the public-facing work is descriptive or predictive without full uncertainty quantification. Our work differs by framing the task as Bayesian inference for latent ability, emphasizing posterior uncertainty and model checking.

2 Preliminary Analysis

We start with a preliminary analysis of the lap time data for 26 clean laps across 20 drivers (2 per team).

Table 1: *Team-Driver Lap Time Stats for 26 Clean laps (in s)*

Team	D1	Mean	SD	Min.	q25, q75	D2	Mean	SD	Min.	q25, q75
RedBull	VER	88.932	1.153	86.993	87.91-89.57	PER	89.163	1.279	87.758	87.984-89.7
Mercedes	HAM	89.656	1.05	88.372	88.81-89.9	RUS	89.526	1.019	88.187	88.69-90.2
Ferrari	LEC	89.331	0.937	88.221	88.55-90.0	SAI	90.522	0.766	89.635	89.85-91.09
McLaren	NOR	89.467	0.942	88.164	88.60-89.87	SAI	89.731	1.172	88.149	88.74-90.59
AstonMartin	ALO	89.742	1.081	88.491	88.62-90.6	STR	89.891	1.129	88.177	88.86-90.85
Alpine	GAS	90.212	0.738	89.307	89.52-90.7	OCO	90.656	0.553	90.113	90.26-90.9
Williams	ALB	90.083	1.442	87.845	88.75-90.7	SAR	90.328	1.338	88.64	88.95-91.4
AlphaTauri	TSU	90.241	0.816	89.256	89.47-90.9	RIC	89.947	1.213	88.571	88.77-90.7
AlfaRomeo	BOT	91.41	0.885	89.998	90.57-92.0	ZHO	90.524	1.030	88.951	89.7-91.2
Haas	MAG	90.968	0.763	89.934	90.44-91.5	SAI	90.689	0.909	89.456	90.09-91.4

Table 1 observes lap-level performance by driver. We observe that while there are differences in mean lap times across drivers and teams, there is also considerable overlap in their distributions, highlighting the need for a more nuanced analysis to disentangle driver ability from other factors.

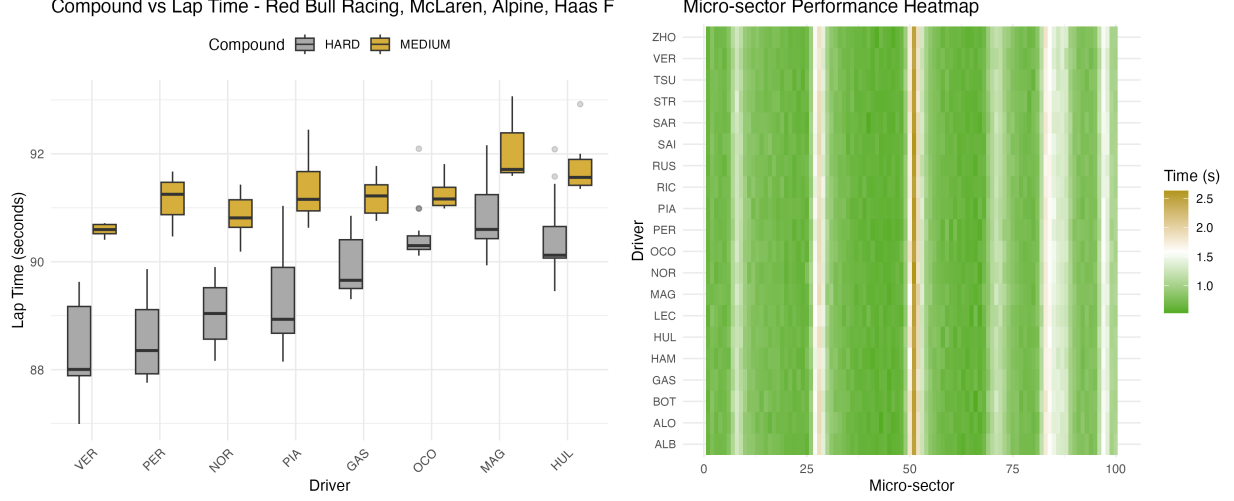


Figure 1.1: Lap time distributions by tire compound and team (every 2 drivers are a single team); Figure 1.2: Heatmap of micro-sector performance across drivers and sectors.

Figure 1.1 shows that lap times are faster with harder compounds, despite comparatively softer compounds generally yielding faster times. This suggests other factors like track temperature, tire degradation and stint length may be influencing performance. We also see an increase trend in lap times across teams, denoting that teams have different performance thresholds. Figure 1.2 reveals patterns in micro-sector performance, with consistent time increase corresponding to turns indicating the importance of reflecting track geometry. For example, sectors 26-31, 49-55, and 80-88 correspond to turns 5, 6-7, and 12-14 respectively (see Figure x.x), and show variability across drivers (hard to distinguish).

3 Methodology

3.1 Modelling

We model micro-sector time y_i (seconds) through its logarithm ($\log(y_i)$) to ensure predictions remain positive on the original scale and to improve approximate normality. We build two models: a baseline (naive) model for interpretability and a full hierarchical model for driver ranking and performance decomposition.

3.1.1 Naive Model

$$\log(y_i) \mid \mu_i \sim \mathcal{N}(\mu_i, \sigma); \quad \sigma \sim \text{Exp}(1)$$

$$\mu_i = \alpha_0 + \alpha_{d[i]} + \beta_1 \text{TyreLife}_i + \beta_2 \text{TrackTemp}_i + \beta_3 \text{Compound}_i + \psi_i \text{TrackStatus}_i$$

3.1.2 Complex Model

3.2

4 Results & Interpretations

4.1 Diagnostics

4.2 Variational Inference

5 Discussion

5.1 Limitations

5.2 Conclusions

6 Appendix

6.1 GitHub

Outputs and code available at [GitHub Repository](#)

NOTE: To ensure no issues in running the code for reproducing the outputs and analysis follow the instructions to setup your environment and run instructions in the `README.md` file.

6.2 Data Information

Variable	Type	Description
Car Telemetry Data across 100 Micro-sectors		
Driver	Categorical	Three-letter driver abbreviation (e.g., VER, HAM)
Team	Categorical	The team or constructor name (e.g., Red Bulls Racing, Mercedes, Ferrari)
LapNumber	Integer (1–58)	Sequential lap index during the race (Race for this session 58 laps long)
micro_sector	Integer (0–99)	Track divided into 100 equal-distance segments
Speed	Continuous (km/h)	Instantaneous vehicle speed at given micro-sector
Throttle	Continuous (0–100)	Throttle application percentage
Brake	Continuous (0–100)	Brake application percentage
RPM	Continuous	Engine revolutions per minute
TyreLife	Integer	Number of laps completed on current tire set
AirTemp	Continuous (°C)	Ambient air temperature
TrackTemp	Continuous (°C)	Track temperature
LapTimeSeconds	Continuous (s)	Total lap time
TimeSeconds	Continuous (s)	Time elapsed from the start of the current micro-sector to its end
Lap-Level Driver Summary Data for 26 Clean Laps		
Driver	Categorical	Three-letter driver abbreviation (e.g., VER, HAM)
Team	Categorical	The team or constructor name (e.g., Red Bulls Racing, Mercedes, Ferrari)
LapNumber	Integer (1–58)	Sequential lap index during the race (Race is 58 long)
LapTimeSeconds	Continuous (s)	Total lap time
Stint	Integer	Stint index between pit stops
Compound	Categorical	Tire compound (SOFT, MEDIUM, HARD)
TrackStatus	Binary	Race continuation status (1=Green, others=flags)
PitInTime	Date-time or NA	Timestamp of pit entry
PitOutTime	Date-time or NA	Timestamp of pit exit

Table 2: *Table Summary of Variables in the Processed Telemetry for 26 Clean Laps*

6.2.1 Processed Datasets

The following are the previews of the two main datasets used in the analysis. The first dataset contains micro-sector telemetry data for 100 segments of each lap, while the second dataset summarizes lap-level information for 26 clean laps.

```
## Rows: 520
## Columns: 12
## $ Driver      <chr> "VER", "VER", "VER", "VER", "VER", "VER", "VER", "VER",~
## $ driver_id   <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ Team        <chr> "Red Bull Racing", "Red Bull Racing", "Red Bull Racing"~
## $ team_id     <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ LapNumber   <dbl> 2, 3, 4, 9, 10, 11, 19, 20, 21, 25, 26, 27, 28, 39, 40,~
## $ Compound    <chr> "MEDIUM", "MEDIUM", "MEDIUM", "MEDIUM", "MEDIUM", "MEDI~
## $ compound_id <dbl> 2, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3~
## $ TyreLife    <dbl> 1, 2, 3, 8, 9, 10, 2, 3, 4, 8, 9, 10, 11, 22, 23, 1, 2,~
## $ Stint       <dbl> 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 3, 3, 3,~
## $ TrackStatus <dbl> 0, 1, 0, 1, 1, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1,~
## $ LapTimeSeconds <dbl> 90.710, 90.408, 90.716, 90.632, 90.506, 90.561, 89.386,~
## $ log_lap_time <dbl> 4.507668, 4.504333, 4.507734, 4.506807, 4.505416, 4.506~
```

Table x.x: Car Telemetry Data across 100 Micro-sectors driven by 20 drivers in 2023 F1 Abu Dhabi Grand Prix

```
## Rows: 52,000
## Columns: 19
## $ Driver      <chr> "VER", "VER", "VER", "VER", "VER", "VER", "VER", "VER",~
## $ driver_id   <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ Team        <chr> "Red Bull Racing", "Red Bull Racing", "Red Bull Racing"~
## $ team_id     <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ compound_id <dbl> 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2~
## $ LapNumber   <dbl> 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2~
## $ micro_sector <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, ~
## $ Speed       <dbl> 233.0000, 248.7500, 262.0000, 268.0000, 271.7500, 245.6~
## $ Throttle    <dbl> 100.00000, 100.00000, 100.00000, 100.00000, 86.75000, 0~
## $ Brake       <dbl> 0, 0, 0, 0, 0, 100, 100, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ rpm         <dbl> 11528.000, 11143.000, 11075.500, 10731.500, 10698.000, ~
## $ AirTemp     <dbl> 27, 27, 27, 27, 27, 27, 27, 27, 27, 27, 27, 27, 27, 27,~
## $ TrackTemp   <dbl> 33.6, 33.6, 33.6, 33.6, 33.6, 33.6, 33.6, 33.6, 33.6, 3~
## $ TyreLife    <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ TrackStatus <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ TimeSeconds <dbl> 0.429, 1.040, 0.720, 0.560, 0.841, 0.760, 1.000, 1.159,~
## $ LapTimeSeconds <dbl> 90.71, 90.71, 90.71, 90.71, 90.71, 90.71, 90.71, 90.71,~
## $ log_time    <dbl> -0.8462983601, 0.0392207132, -0.3285040670, -0.57981849~
## $ log_lap_time <dbl> 4.507668, 4.507668, 4.507668, 4.507668, 4.507668, 4.507~
```

Table x.x: Lap Summary Data of 20 drivers (2 per team) in 2023 F1 Abu Dhabi Grand Prix

7 References

1. Source: Open F1 - The open source API for Formula 1 data <https://openfl.org/>
2. Source: Collected Primary Datasets - https://github.com/sohbatSandhu/fl_telemetry_bayesian_analysis/tree/main/data/main

3. Source: Backup Datasets (1) Kaggle: <https://www.kaggle.com/datasets/gixarde31/f1-2023-abu-dhabi-gp-telemetry-and-lap-data> (2) GitHub repository: https://github.com/sohbatSandhu/f1_telemetry_bayesian_analysis/tree/main/data/backup